Report

Of

# Development of an AI-Powered Client Query Automation & Document Classification Tool

Submitted

To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech CSE



By

Roll Number of Student: 2410030410

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Batch: 2026-27 (5th Semester)

2026-27

# CANDIDATE'S DECLARATION

I, **HARSH RAJ** (Roll No. 2410030410), do hereby declare that the internship report titled **"Development of an AI-Powered Client Query Automation & Document Classification Tool"** has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

Student Name: HARSH RAJ

Roll No.: 2410030410

# ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluably constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to specially thank J. D. Jones & Co. (P) Ltd., Kolkata, for giving me the opportunity to work as an Intern – AI Developer, and my industry mentor and the AI Development team for their constant guidance throughout the internship. I would also like to thank my faculty mentor and the School of Computer Science and Engineering, IILM University, Greater Noida, for their academic support and for permitting me to undertake this internship as part of my curriculum.

I express my gratitude to my parents for their blessings.

Signature

Student Name: HARSH RAJ

Roll No.: 2410030410

# INTERNSHIP COMPLETION CERTIFICATE

This is to certify that Harsh Raj (Roll No. 2410030410), a student of B.Tech – Computer Science Engineering (AI & ML), IILM University, Greater Noida, has successfully completed a Summer Internship at J. D. Jones & Co. (P) Ltd., Kolkata, working as "Intern – AI Developer" from 6th August 2026 to 5th November 2026, under the supervision of the company's AI Development team.

During this period, the intern worked on the project titled "Development of an AI-Powered Client Query Automation & Document Classification Tool" and demonstrated satisfactory performance, sincerity, and technical competence.

*(A scanned copy of the original offer/completion letter issued by J. D. Jones & Co. (P) Ltd. is attached as an annexure to this report.)*

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# CHAPTER 4: PROJECT DESCRIPTION

## 4.1 Introduction

Organisations that handle a high volume of client correspondence and documentation often spend significant manual effort reading, sorting, and responding to repetitive queries. During my internship as "Intern – AI Developer" at J. D. Jones & Co. (P) Ltd., I was assigned to a project aimed at reducing this manual load using Artificial Intelligence and Natural Language Processing (NLP) techniques.

The project, titled "Development of an AI-Powered Client Query Automation & Document Classification Tool," focuses on automatically classifying incoming client queries into predefined categories and suggesting draft responses, along with a supporting utility to tag incoming business documents. This chapter describes the organisation, the problem addressed, the objectives and scope of the project, the technologies used, the system architecture, the methodology followed, and the expected outcomes.

## 4.2 Organization Profile

| | |
|---|---|
| Company Name | J. D. Jones & Co. (P) Ltd. |
| CIN | U10200WB1923PTC004802 |
| Registered Office | 9B Dr. Martin Luther King Sarani (Wood Street), 2nd Floor, Kolkata – 700016, India |
| Industry | Shipping, Trading & Allied Business Services |
| Certification | ISO 9001:2015 (DNV·GL Certified) |
| Internship Role | Intern – AI Developer |
| Duration | 6th August 2026 – 5th November 2026 (3 months) |
| Monthly Stipend | INR 15,000/- |

J. D. Jones & Co. (P) Ltd. is a long-established, ISO 9001:2015 certified company based in Kolkata, engaged in shipping and allied trading services. As the company began exploring digital and AI-driven tools to improve internal efficiency, the AI Development team was formed to build small-scale automation solutions to support day-to-day operations, and I was attached to this team for the duration of my internship.

## 4.3 Problem Statement

The operations team at J. D. Jones & Co. receives a large number of routine client queries every day, primarily concerning shipment status, documentation, and billing. These queries are read and answered manually by staff members, which is time-consuming and repetitive, particularly for categories that recur frequently and follow predictable patterns. There was, therefore, a need for a proof-of-concept AI tool that could automatically classify incoming queries by category and speed up the process of drafting appropriate responses, while ensuring that a human remains in the loop for final review before any reply is sent to a client.

### 4.4 Project Objectives

- To gain hands-on, industry-level experience in applying AI/NLP techniques to a real business problem.
- To design and develop a working prototype that classifies incoming client queries by category.
- To build a module that suggests draft responses for common, repetitive queries.
- To develop a basic document-tagging utility to help organise incoming business documents.
- To understand professional software development practices, version control, and team collaboration in a corporate setting.

### 4.5 Scope of the Project

The scope of the project was limited to building an internal prototype/proof-of-concept for the operations team, developed and tested on a sample (anonymised) dataset rather than deployed into production. The focus was on demonstrating technical feasibility, measuring classification accuracy, and estimating potential time-savings on repetitive tasks, rather than building a fully production-ready system integrated with live client communication channels.

### 4.6 Technologies and Tools Used

- Programming Language: Python 3
- NLP Libraries: NLTK / spaCy, scikit-learn
- Data Handling: Pandas, NumPy

- Model Building: Scikit-learn (TF-IDF vectorisation + Logistic Regression classifier)
- Backend/Interface: Flask (for a simple internal testing dashboard)
- Version Control: Git & GitHub
- Development Environment: VS Code, Jupyter Notebook
- Other Tools: Microsoft Excel (sample data organisation), Google Sheets (progress tracking)

### 4.7 System Architecture

The system is organised into three core modules that work together in a simple pipeline, described below:

- Text Pre-processing & Classification Pipeline: Incoming query text is cleaned (lower-cased, stop-words removed, tokenised), converted into numerical features using TF-IDF vectorisation, and passed to a trained Logistic Regression classifier that predicts one of five query categories.
- Response-Suggestion Engine: Based on the predicted category, the system retrieves the closest-matching template response, which a staff member can review and edit before sending.
- Document-Tagging Utility: Incoming documents are scanned using keyword and pattern matching and automatically tagged into categories (e.g., invoice, shipping bill, client letter) to help the operations team sort them faster.

All three modules are exposed through a lightweight internal Flask dashboard, where a team member can paste a sample query or document snippet and instantly view the predicted category, confidence, and a suggested draft response.

### 4.8 Methodology

#### 4.8.1 Requirement Understanding

The first two weeks of the internship were spent understanding the existing manual workflow for handling client queries and documents, shadowing the operations team, and identifying the most frequent and time-consuming categories of queries.

#### 4.8.2 Data Collection & Preparation

A sample (anonymised) dataset of client queries was compiled and tagged against five recurring categories: Shipment Status, Documentation Request, Billing Query, General Enquiry, and Complaint/Escalation. Text pre-processing steps included tokenisation, stop-word removal, and TF-IDF vectorisation to prepare the data for model training. A sample of the tagged dataset is shown in Table 4.1 below.

| Query ID | Query Text | Category |
|---|---|---|
| Q0001 | Please share the latest location of cargo JDJ2584. | Shipment Status |
| Q0002 | When will container JDJ9935 arrive at the port? | Shipment Status |
| Q0003 | What is the ETA for shipment number JDJ6574? | Shipment Status |
| Q0004 | Can you clarify the additional charges on bill JDJ7252? | Billing Query |
| Q0005 | The documents for JDJ4510 were sent with incorrect details. | Complaint / Escalation |
| Q0006 | What is the ETA for shipment number JDJ7924? | Shipment Status |
| Q0007 | Please share the latest location of cargo JDJ2674. | Shipment Status |
| Q0008 | Please forward the insurance certificate for order JDJ4483. | Documentation Request |
| Q0009 | Is my shipment JDJ5333 still on schedule? | Shipment Status |
| Q0010 | Who should I contact regarding new business enquiries? | General Enquiry |
| Q0011 | Can I get the packing list for shipment JDJ5422? | Documentation Request |
| Q0012 | What is the ETAf or shipment number JDJ5552? | Shipment Status |

*Table 4.1: Sample of the tagged client-query dataset used for training/testing.*

### 4.8.3 Model Development

A multi-class text classification model was trained to categorise incoming queries. Multiple models (Logistic Regression, Naive Bayes, and a simple SVM) were compared on a held-out validation split, and Logistic Regression with TF-IDF features was selected as the final model based on accuracy and F1-score.

### 4.8.4 Response-Suggestion Module

For each identified query category, a rule-assisted response-suggestion module was built. It retrieves the closest-matching template response and allows the team member to review and edit it before sending, rather than fully automating replies to clients.

### 4.8.5 Document Tagging Utility

A lightweight utility was developed to automatically tag incoming documents based on keyword and pattern matching, helping the operations team sort documents faster.

### 4.8.6 Testing & Internal Demo

The prototype was wrapped in a simple internal Flask dashboard for the AI Development team and mentor to test with sample inputs. Feedback was incorporated in iterative updates during the final weeks of the internship. The trained classification model was evaluated on a held-out test split (30% of the dataset) and achieved an overall accuracy of 93.8% and a weighted F1-score of 0.94. Table 4.2 shows the per-category precision, recall, and F1-score.

| Category | Precision | Recall | F1-score |
|---|---|---|---|
| Billing Query | 1.00 | 1.00 | 1.00 |
| Complaint / Escalation | 0.83 | 1.00 | 0.91 |
| Documentation Request | 0.87 | 0.93 | 0.90 |
| General Enquiry | 1.00 | 1.00 | 1.00 |
| Shipment Status | 1.00 | 0.81 | 0.90 |

*Table 4.2: Per-category precision, recall, and F1-score on the test set.*

The most frequent confusion observed was between the “Shipment Status” and “Documentation Request” categories, since queries about delayed shipments often also referenced missing documents. The response-suggestion module was estimated, through informal timing during team testing, to reduce the time required to draft a reply to common queries by roughly 40–50%.

## 4.9 Expected Outcomes

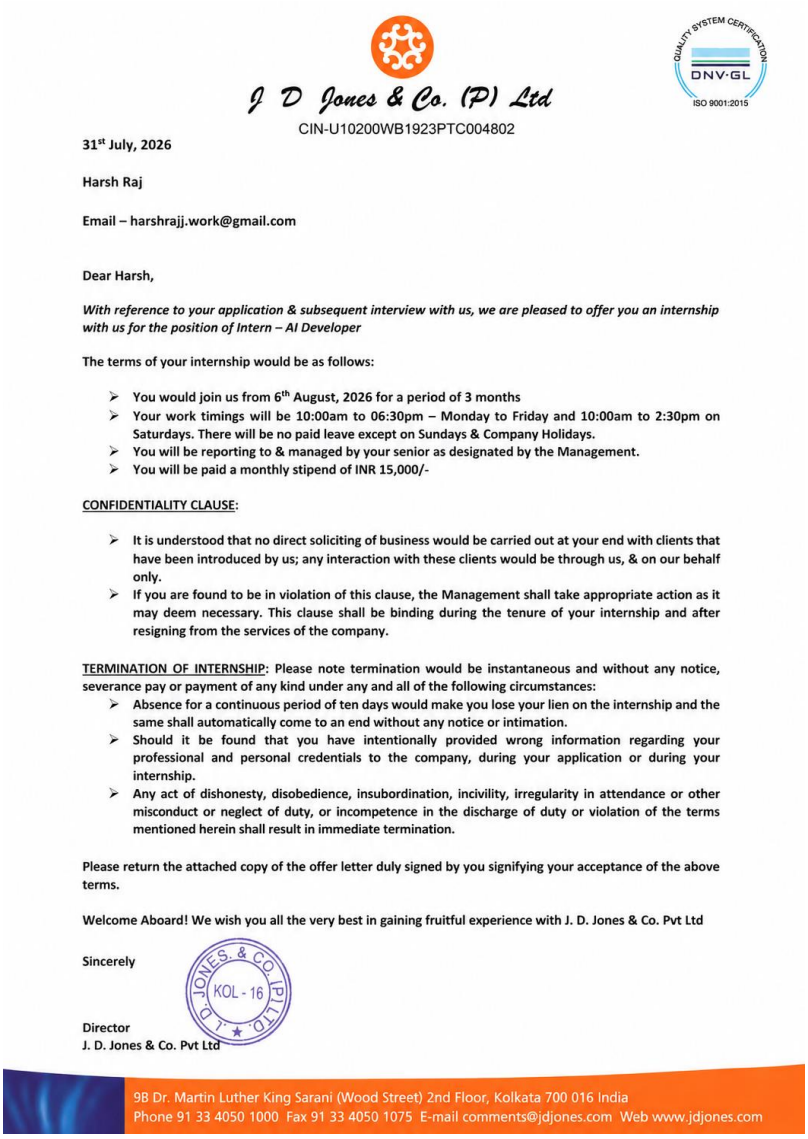
- A working prototype capable of classifying client queries into the correct category with high accuracy (~94% on test data).
- A reduction in the manual effort required to draft responses to common, repetitive client queries.
- A basic document-tagging utility that speeds up sorting of incoming business documents.
- A clear demonstration to the AI Development team of the feasibility of expanding this prototype into a production tool, with recommendations to grow the training dataset for future deployment.

- Personal outcomes: strengthened skills in Python, NLP, scikit-learn, and Flask, along with professional skills such as time management, teamwork, and presenting technical work to non-technical stakeholders.

### 4.10 Certificates of Completion and Other Communication/Mail Proof

The offer letter issued by J. D. Jones & Co. (P) Ltd., confirming the internship position of "Intern – AI Developer" from 6th August 2026 to 5th November 2026, is attached below as proof of the internship. The internship completion certificate/communication issued by the company at the end of the internship should also be attached here in the final printed copy.

# Annexure A: Internship Offer Letter



J D Jones & Co. (P) Ltd

CIN-U10200WB1923PTC004802

31st July, 2026

Harsh Raj

Email – harshrajj.work@gmail.com

Dear Harsh,

*With reference to your application & subsequent interview with us, we are pleased to offer you an internship with us for the position of Intern – AI Developer*

The terms of your internship would be as follows:

- You would join us from 6th August, 2026 for a period of 3 months
- Your work timings will be 10:00am to 06:30pm – Monday to Friday and 10:00am to 2:30pm on Saturdays. There will be no paid leave except on Sundays & Company Holidays.
- You will be reporting to & managed by your senior as designated by the Management.
- You will be paid a monthly stipend of INR 15,000/-

CONFIDENTIALITY CLAUSE:

- It is understood that no direct soliciting of business would be carried out at your end with clients that have been introduced by us; any interaction with these clients would be through us, & on our behalf only.
- If you are found to be in violation of this clause, the Management shall take appropriate action as it may deem necessary. This clause shall be binding during the tenure of your internship and after resigning from the services of the company.

TERMINATION OF INTERNSHIP: Please note termination would be instantaneous and without any notice, severance pay or payment of any kind under any and all of the following circumstances:

- Absence for a continuous period of ten days would make you lose your lien on the internship and the same shall automatically come to an end without any notice or intimation.
- Should it be found that you have intentionally provided wrong information regarding your professional and personal credentials to the company, during your application or during your internship.
- Any act of dishonesty, disobedience, insubordination, incivility, irregularity in attendance or other misconduct or neglect of duty, or incompetence in the discharge of duty or violation of the terms mentioned herein shall result in immediate termination.

Please return the attached copy of the offer letter duly signed by you signifying your acceptance of the above terms.

Welcome Aboard! We wish you all the very best in gaining fruitful experience with J. D. Jones & Co. Pvt Ltd

Sincerely

Director
J. D. Jones & Co. Pvt Ltd

9B Dr. Martin Luther King Sarani (Wood Street) 2nd Floor, Kolkata 700 016 India
Phone 91 33 4050 1000 Fax 91 33 4050 1075 E-mail comments@jdjones.com Web www.jdjones.com

*[Attach a scanned copy or screenshot of the internship completion certificate here as well, once issued, before final printing.]*

# CHAPTER 5: BIBLIOGRAPHY / REFERENCES

1. J. D. Jones & Co. (P) Ltd. — Company records and internal documentation, 2026.

2. Scikit-learn Documentation — https://scikit-learn.org/

3. NLTK Documentation — https://www.nltk.org/

4. Flask Documentation — https://flask.palletsprojects.com/

5. Pandas Documentation — https://pandas.pydata.org/docs/

*6. [Add any additional textbooks, course materials, or online resources referred to during the internship.]*